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Fiber optic cable management trays; assemblies; and methods

Abstract

A telecommunications assembly includes a tray, and telecommunications components. The telecommunications components include fiber routing modules selectively mountable in a plurality of positions to define different fiber pathways. Splice holder modules are used with the fiber routing modules. Attachment features between the tray and modules include a dovetail feature, and crush bumps, biased ends, and defined lengths to protect fibers being managed on the tray.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage Application of PCT/US2020/043396, filed on Jul. 24, 2020, which claims the benefit of U.S. patent application Ser. No. 62/879,314, filed on Jul. 26, 2019, and claims the benefit of U.S. patent application Ser. No. 63/050,619, filed on Jul. 10, 2020, the disclosures of which are incorporated herein by reference in their entireties. To the extent appropriate, a claim of priority is made to each of the above disclosed applications.

TECHNICAL FIELD

(1) The present disclosure relates generally to fiber optic telecommunications cable management trays and components. More particularly, the present disclosure relates to devices, systems and

methods for storing fiber optic cables, such as loops of fiber optic cable slack and components.

BACKGROUND

(2) Telecommunications systems, for example fiber optic closures and other devices, can involve the storage of fiber optic cable components such as splices, splitters, or other devices, and the accompanying fiber optic cable connecting to those components. Slack management is important, especially with respect to splicing. The slack fiber optic cable is needed for accessing the fusion splicing equipment to connect the two fibers. Excessive slack is not a preferred situation in that such slack takes up more space and/or needs to be organized and managed.

(3) In the case of a repair of fiber optic connections, such as a re-splice, additional issues arise with respect to managing the slack. If a repair technician is required to re-route fiber optic cables inside or outside of the closure to locate additional slack to make the repair, such events can be time consuming or costly.

(4) Additionally, the overall size of outside plant closures in general, and repair closures in particular, may be limited. For example, there may be size limits so the closures do not hinder placement in the field. Therefore, if the internal slack storage is limited in space it may limit the technician's ability to properly manage short lengths of spliced fiber required in these smaller closures.

(5) Improvements are desired.

SUMMARY

(6) This disclosure pertains to fiber optic tray systems used by telecommunication closures of the type found in the outside plant. For example, some tray systems utilize fusion splice holder modules to organize and protect spliced optical fibers stored on the trays. These modules have also been integrated with optical components such as WDM and splitters. The trays can also be used in other devices, such as inside plant.

(7) In one example, the CommScope, Inc. FOSC line of splice trays are configured to receive and retain a splice holder module. The trays and modules are provided with compatible attachment features that allow for the modules to be mounted to the trays.

(8) This disclosure also pertains to fiber optic routing of the fiber optic cables in an organized manner within a fiber optic tray or other device where fiber slack needs to be stored relative to the splices.

(9) This disclosure pertains to management of fiber optic slack in a manner where the fibers are managed without going below the minimum bend radius of the respective cable.

(10) This disclosure relates to managing fiber optic cables wherein the slack lengths to be managed may vary between the different fiber optic cables being managed by the fiber optic tray or other device.

(11) This disclosure further pertains to management of fiber optic cable slack in a manner where the fibers are connected to other fibers or devices on both ends, and the slack is managed in loop shapes without having unmanaged segments that can interfere with other cables, get damaged, or create organization and/or use problems for the technician.

(12) This disclosure includes cable management devices provided on a substrate for containing one or more fiber loops, and are mounted to the substrate at a desired spacing to take into account the desired pathways of the stored cable.

(13) In one embodiment, the cable management devices are provided with a base, and management tabs. The cable management device can be constructed with a base, sidewalls, and overhanging tabs. Alternatively, two uprights are on one side of the base, and one upright is on an opposite side of the base. Additional uprights can be added.

(14) The trays, the management devices, and the component modules (for example, splice holders) are provided with compatible attachment features that allow for the management devices and modules to be mounted to the trays. These trays can also include both the cable management devices for managing the fiber loops and the component modules, or those trays can be separate

from the component modules on other trays.

(15) The management devices and component modules are selectively positionable on the substrate to allow for section of desired fiber pathways for the fibers to be managed.

(16) The management devices and component modules can include one or bumps on attachment features to promote a tighter fit with the tray to help reduce or eliminate a gap that could catch a fiber. In one example, the one or more bumps can be provided on a surface of a dovetail of the attachment features which faces toward a base of the management devices and component modules.

(17) The management devices and component modules can include a downward bias on the ends or edges to help reduce or eliminate a gap between the management devices and component modules that could catch a fiber.

(18) The management devices can be provided as long or longer relative to the component modules or other components to help ensure that a fiber does not catch on the edge of the component modules or other component, or to help ensure that a fiber does not go under the component modules or other component.

(19) A variety of additional aspects will be set forth in the description that follows. The aspects can relate to individual features and to combinations of features. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the broad inventive concepts upon which the examples disclosed herein are based.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of the description, illustrate several aspects of the present disclosure. A brief description of the drawings is as follows:

(2) FIG. 1 is a top view of a first embodiment of a fiber optic tray;

(3) FIG. 2 is a perspective view of the fiber optic tray of FIG. 1;

(4) FIG. 3 is a side view of the components mounted to the fiber optic tray of FIG. 1;

(5) FIG. 4 is a bottom view of the components shown in FIG. 3;

(6) FIG. 5 is an exploded perspective view of the fiber optic tray of FIG. 1;

(7) FIG. 6 is a top view like FIG. 1, showing example fiber routing pathways;

(8) FIG. 6A shows example fibers and fiber pathways;

(9) FIG. 6B shows other example fibers and fiber pathways;

(10) FIG. 7 is a top view of a second embodiment of a fiber optic tray;

(11) FIG. 8 is a perspective view of the fiber optic tray of FIG. 7;

(12) FIG. 9 is a side view of the components mounted to the fiber optic tray of FIG. 7;

(13) FIG. 10 is a bottom view of the components shown in FIG. 9;

(14) FIG. 11 is an exploded perspective view of the fiber optic tray of FIG. 7;

(15) FIG. 12 is a top view like FIG. 1, showing different fiber routing pathways;

(16) FIGS. 13-18 are various views of the fiber optic cable routing module used in the trays of FIGS. 1-6, 6A, 6B and 12;

(17) FIGS. 19-24 are various view of the fiber optic cable routing module used in the tray of FIGS. 7-11;

(18) FIGS. 25-30 are various views of another fiber optic cable routing module useable in the trays of FIGS. 1-12;

(19) FIG. 31 is a top view of a third embodiment of a fiber optic tray;

(20) FIG. 32 is an exploded perspective view of the fiber optic tray of FIG. 31;

(21) FIGS. 33-36 are various views of one of the fiber optic cable routing modules shown in the

tray of FIGS. **31** and **32**;

(22) FIGS. **37-46A** are various views of another one of the fiber optic cable routing modules shown in the tray of FIGS. **31** and **32**;

(23) FIGS. **47** and **48** are two views of a further one of the fiber optic cable routing modules shown in the tray of FIGS. **31** and **32**;

(24) FIGS. **49-52** are various views of a still further one of the fiber optic cable routing modules shown in the tray of FIGS. **31** and **32**;

(25) FIGS. **53** and **54** are two views of a still further one of the fiber optic cable routing modules shown in the tray of FIGS. **31** and **32**.

DETAILED DESCRIPTION

(26) Referring to FIGS. **1** to **6**, a telecommunications arrangement **10** is shown in various configurations. In one aspect, the telecommunications arrangement includes a tray **12** for storing telecommunications components and fiber optic cabling. As shown, the tray **12** includes a base wall **14** and a circumferential sidewall **16** to define an interior volume **18**. To retain fiber optic cables within the tray, extension tabs or members **20** can be provided. In the example shown, a plurality of extension tabs **20** are provided extending from the sidewall **16** over the interior volume **18** and in a direction parallel to the base wall **14**. Tray **12** defines hinge pins **40** to allow the tray to be hinged to a support structure, such as a tower, possibly with other trays **12**.

(27) The tray **12** is also shown as including a plurality of attachment arrangements **30** arranged in an array extending the length of the tray **12** along the base wall **14**. The attachment arrangements **30** are configured to receive and retain various telecommunications components or devices, such as fiber routing modules, fiber optic adapters, splice holders, optical splitters, wavelength division multiplexers (WDM), and the like that are provided with an attachment arrangement compatible with the attachment arrangements **30**. As shown, a splice holder device **140** and a fiber routing device **160** are used in various selectable positions to define selectable fiber routing pathways.

(28) In one aspect, each attachment arrangement **30** of the tray includes a pair of connection points **32**, **34**. The first connection point **32** is configured as a t-shaped opening in the base wall **14**. The second connection point **34**, arranged oppositely from the first connection point **32**, is configured with a first opening **34a** into which a cantilevered tab **34b** extends. Similarly constructed attachment arrangements for use with interconnecting telecommunications components are shown and described in Patent Cooperation Treaty (PCT) Application Serial Number PCT/US2019/17904, filed on Feb. 13, 2019, the entirety of which is incorporated by reference herein; Patent Cooperation Treaty (PCT) Application Serial Number PCT/US2019/028245, filed on Apr. 19, 2019, the entirety of which is incorporated by reference herein; and U.S. Provisional Patent Application Ser. No. 62/824,824, filed on Mar. 27, 2019, the entirety of which is incorporated by reference herein.

(29) The attachment arrangements **30** are arranged in a linear array in the example tray **12**. An example pitch of 7.7 mm is shown. Not all attachment arrangements **30** may be used. Some devices, such as splice holders **140** span multiple arrangements **30**.

(30) In one aspect, the attachment arrangement **110** on each device **140**, **160** is configured with a pair of attachment features **112**, each including a pair of oppositely arranged dovetail structures **114** and ramp structures **116**. Accordingly, the main body **102** can be attached to a pair of attachment arrangements **30** of the tray **12**. It is also noted that the pairs of dovetail and ramped structures **114**, **116** are symmetrically arranged such that the device is symmetrical about a longitudinal axis X. With such a configuration, the devices **140**, **160** can be easily mounted to the tray **12** in either orientation.

(31) As shown, each of the ramp structures **116** includes a ramped surface **116a** and a stop surface **116b**. As configured, the attachment arrangement **110** can be connected to the attachment arrangement **30** by aligning the main body bottom side **102b** with the base wall **14** such that the dovetail structures **114** and ramp structures **116** drop into the openings **32a**, **32b** of the connection

point **32** and the opening **34a** of the connection point **34**. From this position, the main body **102** can be displaced laterally in a direction towards the cantilevered tab **34b** until the tab snaps over the ramped surface **116a** and abuts the stop surface **116b**. In this position, the connection arrangements **110**, **42** are fully interconnected. The U.S. Ser. No. 62/824,824 application describes a generally similar connection arrangement between two components.

(32) As shown in the Figures, a tray **12** is shown with a plurality of fiber optic splice holder modules **140** mounted to tray **12**. A plurality of fiber routing modules **160** are also shown mounted to the tray. The fiber routing modules can be positioned as desired on tray **12** for managing fiber optic cables extending to and from the splices of the splice holders **140**. Fiber routing modules **160** have generally planar exterior side walls or side projections. An inner trough area is constructed to retain one or more fibers passing through the interior passage. Similarly, splice holders **140** have generally planar vertical exterior sidewalls or side projections to allow for side by side placement with other splice holder modules **140** or fiber routing modules **160**. As shown in the Figures, the fiber routing modules can have a variety of constructions for the fiber retention features.

(33) The tray **12** is shown with either splice pathways or fiber slack pathway along the full middle length between pathways A and C.

(34) Other telecommunications components can be mounted to tray **12** including splitter modules. The splitter modules can be mounted instead of splice holder modules **140** or in combination with splice holder modules **140**. The fiber routing modules **160** can be used to guide fibers to and from the splice holder modules **140**, and the splitter modules, or other components on tray **12**. Different cable bundles can be segregated by using different routing modules **160**.

(35) The attachment arrangements **30** in the form of the dovetail and corresponding slots allows for flexibility in the arrangement of the components on tray **12**. Trays can be customized for different fiber arrangements depending on the nature of the components stored on the tray. Further, the locations of the fiber routing modules **160** can be selected upon desired locations for managing the different fiber loops between incoming and outgoing fibers.

(36) FIG. **6** shows various routing pathways. Cables enter and exit tray **12** at ports **142**, **144**. The cables can pass around the exterior of a middle region defined by pathways A, B, C, D.

(37) Middle region **152** receives individual fibers and components. Therefore, there is a need to manage various fibers without having to always be restricted to pathways A, B, C, D. Instead, as shown in FIG. **6** additional pathways E, F, and G are provided in the illustrated example to facilitate additional fiber management with respect to telecommunications devices **140a**, **140b**, **140c**, **140d**.

(38) FIG. **6A** shows a fiber **1** and a fiber **2** entering and exiting tray **12** and being connected at a splice on splice holder **140a**. Fiber **1** can be tied off at **50a** and fiber **2** can be tied off at **50b**. The various pathways are utilized to store the fiber lengths necessary to be managed by tray **12**. Additional loops can be managed by tray **12**. In the example of FIG. **6A**, only a few loops are shown for illustration purposes. Additional loops and/or different pathways can be utilized.

(39) FIG. **6B** shows a different fiber routing arrangement for fiber and fiber **4**. These fibers are spliced on splice holder **140c**.

(40) Referring now to FIG. **7**, tray **12** is shown with different splice holders **240**, and different fiber routing modules **260**. However, the concepts of FIGS. **1-6**, **6A**, **6B** are similar in that various fiber routing pathways can be selected and organized as desired.

(41) Referring now to FIG. **12**, tray **12** includes a different arrangement of fiber splice holders **140**, and fiber routing modules **160**. As shown, two splice modules **140** are shown mounted side-by-side and two fiber routing modules **160** are shown mounted side-by-side. FIG. **12** illustrates the versatility of using the attachment arrangements **30** with appropriately sized modules and components to make for efficient storage and efficient cable routing.

(42) FIGS. **13-18** show fiber routing module **160** having a base **166**, sidewalls **164**, overhanging tabs **162**, and a fiber passage **168**. Each splice holder **140** includes a plurality of channels **142**

defined by a base **144**, sidewalls **146**, and inner dividers (see FIG. **3**). Splice holder **240** is constructed similarly.

(43) Referring now to FIGS. **19-24**, fiber routing module **260** is shown in greater detail including a base **266**, sidewalls **264**, tabs **262**, and defining a passage **268**.

(44) Referring now to FIGS. **25-30**, a fiber routing module **360** is shown with a base **366**, sidewalls **364**, tabs **362**, fiber guides **402**, **404**, which together define a passage for fiber.

(45) Referring to FIGS. **31** and **32**, a telecommunications arrangement **510** is shown in various configurations. In one aspect, the telecommunications arrangement includes a tray **512** for storing telecommunications components and fiber optic cabling. As shown, the tray **512** includes a base wall **514** and a circumferential sidewall **516** to define an interior volume **518**. To retain fiber optic cables within the tray, extension tabs or members **520** can be provided. In the example shown, a plurality of extension tabs **520** are provided extending from the sidewall **516** over the interior volume **18** and in a direction parallel to the base wall **514**. Tray **512** defines a hinge pin **540** to allow the tray to be hinged to a support structure, such as a tower, possibly with other trays **512**.

(46) The tray **512** is similar in construction as tray **10**, and is also shown as including a plurality of attachment arrangements **30** arranged in an array extending the length of the tray **512** along the base wall **514**. The attachment arrangements **30** are configured to receive and retain various telecommunications components or devices, such as fiber routing modules, fiber optic adapters, splice holders, optical splitters, wavelength division multiplexers (WDM), and the like that are provided with an attachment arrangement compatible with the attachment arrangements **30**. As shown, a splice holder device **140** and example of various fiber routing devices **560**, **660**, **760**, **860**, and **960** are useable in various combinations and in various selectable positions to define selectable fiber routing pathways. Devices **160**, **260**, **360** can also be used on tray **512** as desired, including the features noted below.

(47) FIGS. **33-36** are various views of one of the fiber optic cable routing modules **560** shown in the tray of FIGS. **31** and **32**.

(48) FIGS. **37-46A** are various views of another one of the fiber optic cable routing modules **660** shown in the tray of FIGS. **31** and **32**.

(49) FIGS. **47** and **48** are two views of a further one of the fiber optic cable routing modules **760** shown in the tray of FIGS. **31** and **32**.

(50) FIGS. **49-52** are various views of a still further one of the fiber optic cable routing modules **860** shown in the tray of FIGS. **31** and **32**.

(51) FIGS. **53** and **54** are two views of a still further one of the fiber optic cable routing modules **960** shown in the tray of FIGS. **31** and **32**.

(52) In one aspect, the attachment arrangement **110** on each device **140**, **560**, **660**, **760**, **860**, and **960** is similar to that described above. One difference is shown with respect to the fiber routing devices **660**, **860** and **960**. The fiber routing devices **660**, **860** and **960** can include one or bumps **562** on angled surfaces **564** of the dovetail structures **114**. The bumps **562** on the attachment features to the tray **512** promote a tighter fit with the tray **512** to help reduce or eliminate a gap that might form due to tolerances from manufacture, and that gap could catch (trap) a fiber. In one example, the one or more bumps **562** can be provided on each surface **564** of the dovetail **114** of the attachment features which faces toward a base **568** of the management devices, and the component modules, e.g. splice holders. Each bump **562** can engage an angled surface **570** of the openings **32**, **34** of tray **512**. Bumps **562** and the respective angled surfaces **564**, **570** draw the two parts closer together. Bumps **562** can function as crush ribs which allow the parts to engage one another and deform as needed to create the tighter fit without preventing the parts from being put together at all due to a tolerance misfit. See FIG. **45**. Bumps **562** can take the form of partial spheres as shown. Other shapes include laterally extending, linear ribs extending parallel to the tray, or sharper pyramids or points. Shapes that are easily molded are preferred, but not required.

(53) The management devices and component modules can also include a downward bias on the

ends or edges to help reduce or eliminate a gap between the management devices and component modules and the tray **512** that could catch a fiber. Devices **660** and **860** include ends **662** that are bendable upward around areas **666** so as to stay engaged with the top surface **664** of the tray **512**. See FIGS. **46** and **46A**. The bendable ends **662** do not affect the middle areas where the attachment arrangement **110** is located.

(54) As shown in the FIGS. **31** and **32**, tray **512** is shown with a plurality of fiber optic splice holder modules **140** mounted to tray **512**, as examples of just one arrangement. Any combination of plurality of fiber routing modules noted above can be used as desired. The fiber routing modules can be positioned as desired on tray **512** for managing fiber optic cables extending to and from the splices of the splice holders **140** or any other devices on tray **512**.

(55) Fiber routing modules can have one generally planar exterior side wall **762**, a base **764**, and top projections or tabs **766** over the base (see modules **560**, **660**); or side projections **782** in two rows extending from a base **784**, and each with a top projection or tab **786** over the base (see modules **760**, **860**, and **960**). Two sidewalls **762** can be provided if desired. Single rows of side projections **782** can be provided if desired. An inner trough area is constructed to retain one or more fibers passing through the interior passage.

(56) The management devices **160**, **260**, **360**, **560**, **660**, **760**, **860**, and **960** can be provided in a variety of lengths. In one example, **560**, **660**, **760**, **860** can be as long or longer relative to the component modules **140** or other components to help ensure that a fiber does not catch on the edge of the component modules or other component, or to help ensure that a fiber does not go under the component modules or other component. See FIG. **31**.

(57) The various examples described above are provided by way of illustration only and should not be construed to limit the scope of the present disclosure. Those skilled in the art will readily recognize various modifications and changes that may be made with respect to the examples and applications illustrated and described herein without departing from the true spirit and scope of the present disclosure.

Claims

1. A telecommunications assembly comprising: a tray defining a base wall and a side wall extending from the base wall; a splice holder module mounted to the base wall in one of a plurality of selectable positions laterally on the base wall; and a fiber routing module mounted to the base wall in a side-by-side arrangement with the splice holder module in one of a plurality of selectable positions laterally on the base wall, wherein the fiber routing module and the splice holder module have upright planar walls that face each other, and wherein the fiber routing module includes an inner passageway with at least one overhanging tab; wherein a plurality of splice holder modules and a plurality of fiber routing modules are provided mounted to the tray; wherein each of the plurality of splice holder modules and each of the plurality of fiber routing modules include a dovetail mounting feature for connecting to the base wall of the tray; wherein each of the plurality of splice holder modules and each of the plurality of fiber routing modules are selectably mountable in a plurality of different locations on the base wall to define a plurality of different fiber routing pathways.

2. The telecommunications assembly of claim 1, wherein each splice holder module and each fiber routing module include one or more bumps on an attachment feature to promote a tighter fit with the base wall of the tray to help reduce or eliminate a gap that could catch a fiber, wherein the one or more bumps are provided on a surface of a dovetail of the attachment feature which faces toward a base of the splice holder module or the fiber routing module.

3. The telecommunications assembly of claim 1, wherein each splice holder module and each fiber routing module include a downward bias on downward facing ends or edges to help reduce or eliminate a gap between the base wall of the tray and the splice holder module or the fiber routing

module that could catch a fiber.

4. The telecommunications assembly of claim 1, wherein each fiber routing module is as long or longer relative to an adjacent splice holder module to help ensure that a fiber does not catch on an edge of the splice holder module, or to help ensure that the fiber does not go under the splice holder module or the fiber routing module or other component.
5. The telecommunications assembly of claim 1, wherein the upright planar walls of the fiber routing module are comprised of at least one side wall projection extending into the at least one overhanging tab, wherein one or more openings in the upright planar walls of the fiber routing module are defined between two adjacent side wall projections.
6. The telecommunications assembly of claim 1, wherein the fiber routing module includes a single upright planar wall positioned along one side of the fiber routing module, with the opposite side of the fiber routing module being open.
7. A telecommunications assembly comprising: a tray defining a base wall and a side wall extending from the base wall; a splice holder module mounted to the base wall in one of a plurality of selectable positions laterally on the base wall; and a fiber routing module mounted to the base wall in a side-by-side arrangement with the splice holder module, in one of a plurality of selectable positions laterally on the base wall, wherein the fiber routing module is as long or longer relative to the splice holder module to help ensure that a fiber does not catch on an edge of the splice holder module, or to help ensure that the fiber does not go under the splice holder module; wherein the splice holder module and the fiber routing module each include one or more bumps on an attachment feature to promote a tighter fit with the base wall of the tray to help reduce or eliminate a gap that could catch the fiber, wherein the one or more bumps are provided on a surface of a dovetail of the attachment feature which faces toward a base of the splice holder module or the fiber routing module.
8. The telecommunications assembly of claim 7, wherein each of the splice holder module and the fiber routing module includes a dovetail mounting feature for connecting to the base wall of the tray.
9. The telecommunications assembly of claim 7, wherein a plurality of splice holder modules and fiber routing modules are selectably mountable in a plurality of locations to define different fiber routing pathways.
10. The telecommunications assembly of claim 7, wherein the splice holder module and the fiber routing module each include a downward bias on downward facing ends or edges to help reduce or eliminate a gap between the base wall of the tray and the splice holder module or the fiber routing module that could catch the fiber.
11. A telecommunications assembly comprising: a tray defining a base wall and a side wall extending from the base wall; a splice holder module mounted to the base wall; a fiber routing module mounted to the base wall, wherein the fiber routing module includes an inner passageway with at least one overhanging tab; wherein each of the splice holder module and the fiber routing module includes a dovetail mounting feature for connecting to the base wall of the tray; and wherein the splice holder module and/or the fiber routing module includes one or more bumps on an attachment feature to promote a tighter fit with the base wall of the tray to help reduce or eliminate a gap that could catch a fiber, wherein the one or more bumps are provided on a surface of the dovetail mounting feature which faces toward a base of the splice holder module or the fiber routing module.
12. The telecommunications assembly of claim 11, wherein a plurality of splice holder modules and a plurality of fiber routing modules are selectably mountable in a plurality of locations to define different fiber routing pathways.
13. The telecommunications assembly of claim 11, wherein the splice holder module or the fiber routing module include a downward bias on ends or edges to help reduce or eliminate the gap between the base wall of the tray and the splice holder module or the fiber routing module that

could catch the fiber.

14. A telecommunications assembly comprising: a tray defining a base wall and a side wall extending from the base wall, the base wall having an upper surface; a splice holder module mounted to the base wall, the splice holder module including a lower surface engaged with the upper surface of the base wall; a fiber routing module mounted to the base wall, wherein the fiber routing module includes an inner passageway with at least one overhanging tab, the fiber routing module including a lower surface engaged with the upper surface of the base wall; and wherein the splice holder module and the fiber routing module each include a downward bias on opposite ends to help reduce or eliminate a gap between the upper surface of the base wall of the tray and the lower surface of the splice holder module and the lower surface of the fiber routing module that could catch a fiber; wherein the fiber routing module includes a plurality of internal fiber guides restricting the inner passageway.

15. The telecommunications assembly of claim 14, wherein a plurality of splice holder modules and a plurality of fiber routing modules are selectably mountable in a plurality of locations to define different fiber routing pathways.

16. A telecommunications assembly comprising: a tray defining a base wall and a side wall extending from the base wall; a splice holder module mounted to the base wall in one of a plurality of selectable positions laterally on the base wall; and a fiber routing module mounted to the base wall in a side-by-side arrangement with the splice holder module in one of a plurality of selectable positions laterally on the base wall, wherein the fiber routing module and the splice holder module have upright planar walls that face each other, and wherein the fiber routing module includes an inner passageway with at least one overhanging tab; wherein the splice holder module or the fiber routing module include one or more bumps on an attachment feature to promote a tighter fit with the base wall of the tray to help reduce or eliminate a gap that could catch a fiber, wherein the one or more bumps are provided on a surface of a dovetail of the attachment feature which faces toward a base of the splice holder module or the fiber routing module.
